Task 1:

Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Answer:

Steps to Install and Set Up DNS Server

1. Open the browser and visit the official website: https://simpledns.plus/
2. Download and install Simple DNS Plus on your Windows computer.
3. Open the Simple DNS Plus application.
4. Go to the Zones section.
5. Click on Add New Zone or the option to create a new zone.
6. Select Primary Zone.
7. Enter the domain name musicmania.in.
8. Save the zone configuration.
9. Add a new DNS record of type A Record.
10. Enter the hostname as www and provide your computer's IPv4 address.
11. Save the DNS record.
12. Ensure that the DNS server service is running.
13. Open Command Prompt.
14. Run the command ipconfig to find your computer's IPv4 address.
15. Test the DNS configuration using the command:

nslookup www.musicmania.in <DNS_Server_IP>

16. Verify that the domain resolves to the configured IP address.

Result: The DNS server was installed and configured, and a new DNS zone named musicmania.in was created and tested.

Task 2:

Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Answer:

Steps Performed

1. I opened the Simple DNS Plus application on my computer.
2. I opened the DNS zone named musicmania.in.
3. I selected the option to add a new DNS record.
4. I selected A Record as the record type.
5. I entered www in the hostname field.
6. I entered 192.168.1.50 as the IPv4 address.
7. I saved the DNS record.
8. I checked that the A record was added to the musicmania.in zone.
9. I opened Command Prompt on my computer.
10. I entered the command nslookup www.musicmania.in to check the DNS mapping.
11. I checked the output to verify whether www.musicmania.in resolved to 192.168.1.50.

Result: An A record was configured to map www.musicmania.in to the IP address 192.168.1.50, and the mapping was tested using nslookup.

Task 3:

Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Steps Performed

1. I opened the Simple DNS Plus application.
2. I selected the musicmania.in DNS zone.
3. I added a new DNS record.
4. I selected CNAME Record as the record type.
5. I entered play as the hostname.
6. I entered www.musicmania.in as the canonical hostname.
7. I saved the CNAME record.
8. I opened Command Prompt.
9. I executed the command nslookup play.musicmania.in.
10. I checked whether play.musicmania.in resolved to the IP address 192.168.1.50.

Result: A CNAME record was configured to point play.musicmania.in to www.musicmania.in. The DNS lookup should resolve to 192.168.1.50 when the DNS server is configured correctly.

Task 4:

Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Answer:

Steps Performed

1. I checked the DNS zone configuration and verified that the domain name musicmania.in was entered correctly.
2. I checked the DNS configuration files to ensure that the zone and DNS records were configured properly.
3. I used the named-checkconf command to check the BIND configuration for errors.
4. I used the named-checkzone command to verify the configuration of the musicmania.in zone.
5. I checked whether the DNS service was running using the command:

sudo systemctl status bind9

6. I restarted the DNS service using the command:

sudo systemctl restart bind9

7. I checked the DNS resolution using the nslookup command:

nslookup musicmania.in

8. I used the dig command to get detailed information about the DNS response:

dig musicmania.in

9. I checked the DNS records and verified that the A record and other required records contained the correct values.
10. If the domain still did not resolve, I checked the DNS server IP address, firewall settings, and DNS port 53 connectivity.

Result: I documented the DNS troubleshooting process using configuration checks, service status and restart commands, and DNS testing commands to identify and resolve domain name resolution problems.